5, 28, 19, 15, 20, 33, 12, 17, 10

1) $h(k) = k \bmod 9$
$m = 9$ - chain.

0	1	2	3	4	5	6	7	8
	28	20	12		5	15		17
	19					33		
	10							

5: $5 \% 9 = 5$
28: $28 \% 9 = 1$
19: $19 \% 9 = 1$
15: $15 \% 9 = 6$
20: $20 \% 9 = 2$
33: $33 \% 9 = 6$
12: $12 \% 9 = 3$
17: $17 \% 9 = 8$
10: $10 \% 9 = 1$

2) $m = 1{,}000$ $\quad h(k) = \lfloor m(kA \bmod 1) \rfloor$, $\quad A = \frac{1}{2}(\sqrt{5} - 1) \approx 0.618$

Keys: 61, 62, 63, 64, 65

61: $kA = 61(0.618) = 37.7 \rightarrow \bmod 1 = \lfloor 0.7 \rightarrow \times 1000 \rfloor = 700$ (700.07)
62: $kA = 62(0.61803) = 38.31786 \% 1 = 0.31786 \times 1000 = \lfloor 317.86 \rfloor = 317$
63: $\times 0.61803 = (38.93589) \bmod 1 = .93589 \times 1000 = \lfloor 935.89 \rfloor = 935$
64: $\times 0.61803 = 39.55392 \% 1 = 0.55392 \times 1000 = \lfloor 553.92 \rfloor = 553$
65: $\times 0.61803 = 40.17195 \% 1 = 0.17195 \times 1000 = \lfloor 171.95 \rfloor = 171$
*

61 → 700
62 → 317
63 → 935
64 → 553
65 → 171

3) 10, 22, 31, 4, 15, 28, 17, 88, 59

$m = 11$ $h'(k) = k$

recall $h(k,i) = (h'(k) + i) \bmod m$

a) Linear Probing

0	1	2	3	4	5	6	7	8	9	10
22	88			4	15	28	17	59	31	10

$i=0$

10: $(k+0) \bmod 11 = 10 \bmod 11 = 10$ ✓

22: $(k+0) \bmod 11 = 22 \bmod 11 = 0$ ✓

31: $(k+0) \bmod 11 = 31 \bmod 11 = 9$ ✓

4: $(k+0) \bmod 11 = 4 \bmod 11 = 4$ ✓

15: $(k+0) \bmod 11 = 15 \bmod 11 = 4$ - collision $i{+}{+}$ $(k+1)\%11 = 16\%11 = 5$ ✓

$i=0$ 28: $28\%11 = 6$ ✓

17: $17\%11 = 6$ - collision $i{+}{+}$, $(17+1)\%11 = 18\%11 = 7$ ✓

$i=0$ 88 $88\%11 = 0$ - collision $i{+}{+}$, $(88+1)\%11 = 89\%11 = 1$ ✓

$i=0$ 59 $59\%11 = 4$ - collision $i{+}{+}$, $(59+1)\%11 = 60\%11 = 5$ ✗

collision #2 $i{+}{+}$, $(59+2)\%11 = 61\%11 = 6$ ✗

collision #3 $i{+}{+}$, $(59+3)\%11 = 62\%11 = 7$ ✗

collision #4 $i{+}{+}$, $(59+4)\%11 = 63\%11 = 8$ ✓

$i =$

$h_1(k) = k$

$h_2(k) = 1 + (k \bmod (m-1))$

$h(k,i) = [k + i(1 + (k \bmod (m-1)))] \bmod m$

c) Double Hashing

0	1	2	3	4	5	6	7	8	9	10
22		59	17	4	15	28	88		31	10

10: $[10 + 0] \bmod 11 = 10$ ✓

22: $22 \% 11 = 0$ ✓

31: $31 \% 11 = 9$ ✓

4 $4 \% 11 = 4$ ✓

15 $15 \% 11 = 4$ col i++ $[15 + (1 + 15 \bmod 10)] \bmod 11 = 21 \% 11 = 10$ ✗

i++ col $[15 + 2(1 + 15 \bmod 10)] \bmod 11 = 27 \% 11 = 5$ ✓

i=0 28: $28 \% 11 = 6$ ✓

17: $17 \% 11 = 6$ col i++ $[17 + (1 + 17 \bmod 10)] \bmod 11 = 25 \% 11 = 3$ ✓

i=0 88 $88 \% 11 = 0$ col i++ $[88 + (1 + 88 \bmod 10)] \bmod 11 = 9$, $97 \% 11$ ✗

col i++ $[88 + 2(1 + 88 \bmod 10)] \bmod 11 = 106 \% 11 = 7$ ✓

i=0 59 $59 \% 11 = 4$ col i++ $[59 + (1 + 59 \bmod 10)] \bmod 11 = 69 \% 11 = 3$ ✗

col i++ $[59 + 2(1 + 59 \bmod 10)] \bmod 11 = 79 \% 11 = 2$ ✓

Note: terrible c_1 & c_2 for this data
<u>much</u> better would be $c_1=0$ $c_2=1$

10, 22, 31, 4, 15, 28, 17, 88, 59

3b Quadratic Probe

$c_1=1$, $c_2=3$ $\quad h'(k)=k$ $\quad$ ~~$h(k,i)=(k+i^2) \bmod m$~~

$\longrightarrow h(k,i)=(k+i+3i^2) \bmod m$

0	1	2	3	4	5	6	7	8	9	10
22		88	17	4		28	59	15	31	10

10: $(10+0+0) \bmod 11 = 10\%11 = 10$ ✓

22: $22\%11 = 0$ ✓

31: $31\%11 = 9$ ✓

4: $4\%11 = 4$ ✓

15: $15\%11 = 4$ col i++ $(15+1+3(1)^2)\%11 = 19\%11 = 8$ ✓

i=0 28: $28\%11 = 6$ ✓

17: $17\%11 = 6$ col i++ $(17+1+3(1)^2)\%11 = 21\%11 = 10$ col i++ $(17+2+3(2)^2)\%11 =$

col, i++ $(17+3+3(3)^2)\%11 = (47\%11) = 3$ ✓ $\quad 31\%11 = 9$

i=0 88: $88\%11 = 0$

<u>collision</u> i++, $(88+1+3(1)^2)\%11 = 92\%11 = 4$

collision i++, $(88+2+3(2)^2)\%11 = 102\%11 = 3$

collision i++, $(88+3+3(3)^2)\%11 = 118\%11 = 8$

collision i++, $(88+4+3(4)^2)\%11 = 140\%11 = 8$

collision i++, $(88+5+3(5)^2)\%11 = 168\%11 = 3$

collision i++, $(88+6+3(6)^2)\%11 = 202\%11 = 4$

collision i++, $(88+7+3(7)^2)\%11 = 242\%11 = 0$

collision i++, $(88+8+3(8)^2)\%11 = 288\%11 = 2$ ✓

i=0 59) $59\%11 = 4$ <u>COL</u> i++ $(59+1+3(1)^2) = 63\%11 = 8$ ✗

<u>COL</u> i++ $(59+2+3(2)^2) = 73\%11 = 7$ ✓

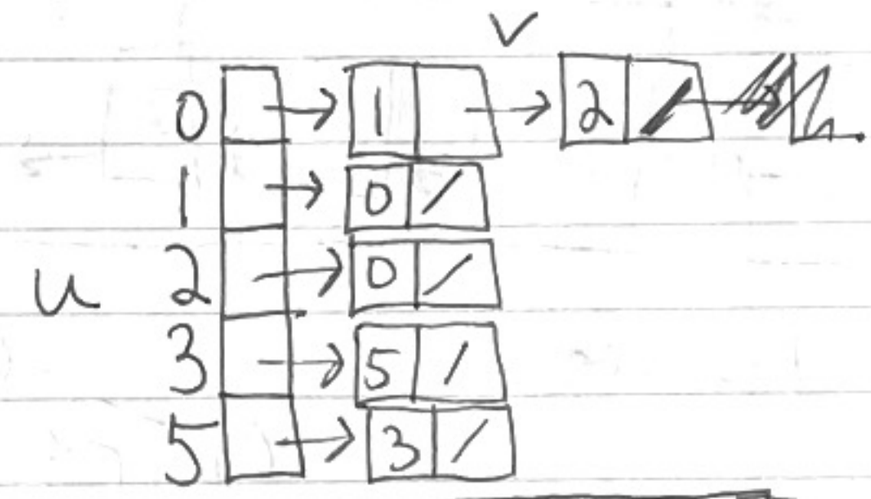
4)

	{a}	{b}	{c}	{d}	{e}	{f}	{g}	{h}	{i}	{j}	{k}
(d, i)	{a}	{b}	{c}	{d, i}	{e}	{f}	{g}	{h}		{j}	{k}
(f, k)	{a}	{b}	{c}	{d, i}	{e}	{f, k}	{g}	{h}		{j}	
(g, i)	{a}	{b}	{c}	{d, g, i}	{e}	{f, k}		{h}		{j}	
(b, g)	{a}		{c}	{d, b, g, i}		{f, k}	{e}	{h}		{j}	
(a, h)			{c}	{d, b, g, i}		{f, k}	{e}	{a, h}		{j}	
(i, j)			{c}	{d, b, g, i, j}		{f, k}	{e}	{a, h}			
(d, k)			{c}	{d, b, g, f, k, i, j}			{e}	{a, h}			
(b, j)			{c}	{d, b, g, f, k, i, j}			{e}	{a, h}			
(d, f)			{c}	{d, b, g, f, k, i, j}			{e}	{a, h}			
(g, j)			{c}	{d, b, g, f, k, i, j}			{e}	{a, h}			
(a, e)			{c}	{d, b, g, f, k, i, j}				{a, e, h}			
(i, d)	{a, e, h}			{b, d, f, g, i, j, k}				{c}			

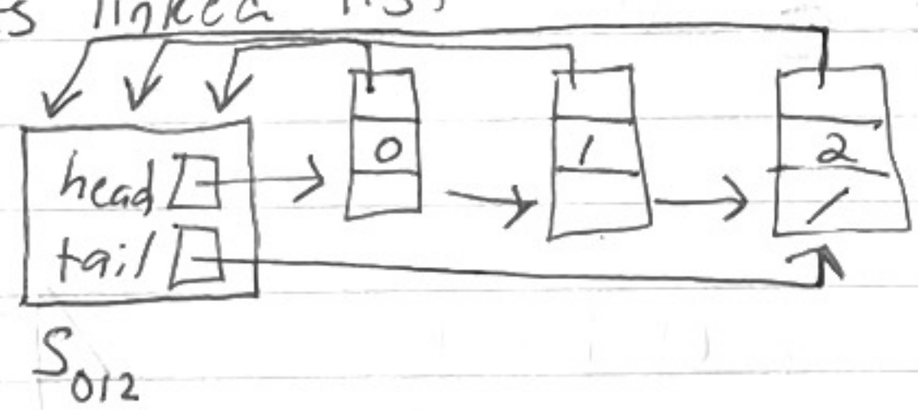
- reordered to read

5a)

=



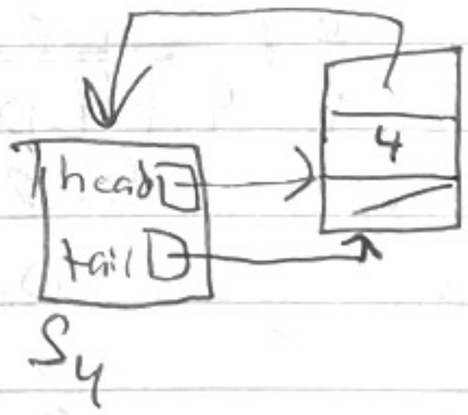
As linked list



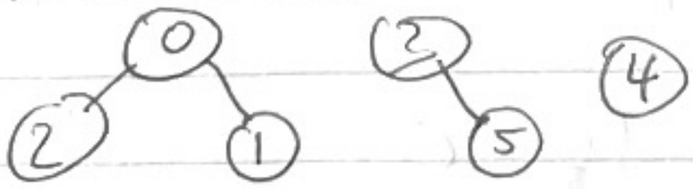
b) Make-set (4)

i) listed weight heuristic

these two (S_{012}, S_{35}) as above, and

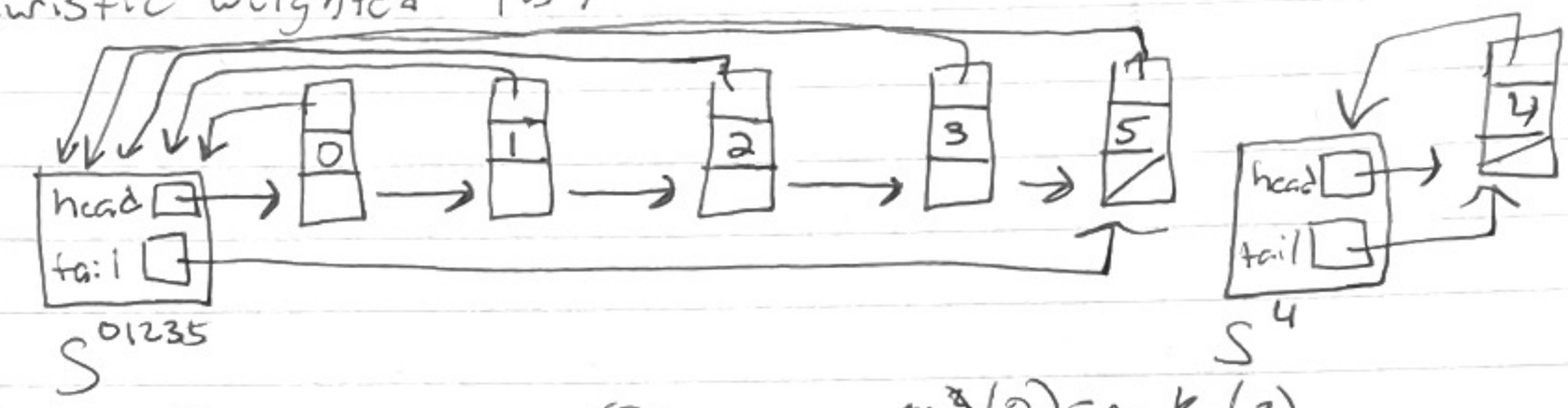


ii) rank tree

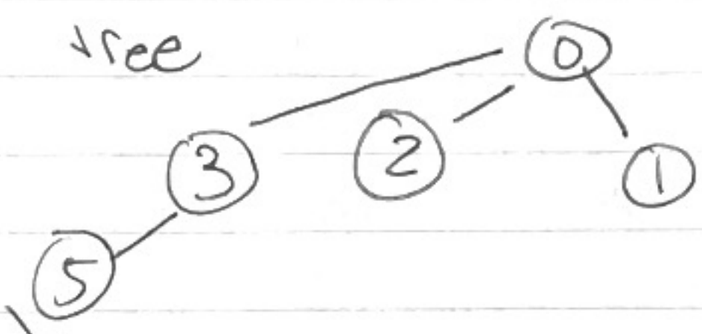


b.2) UNION(1,5)

i) heuristic weighted list



ii) ranked tree

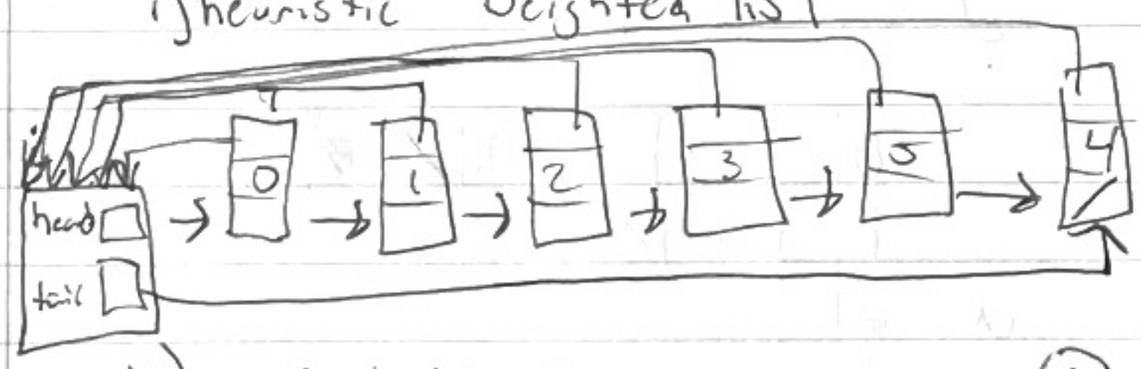


rank(0) = rank(3)

rank(0) ++

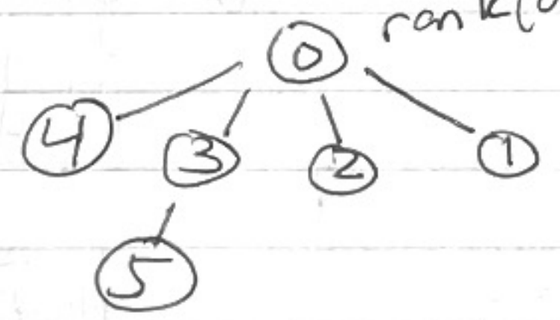
b3) Union(4,5)

i) heuristic weighted list



ii) ranked tree

rank(0) > rank(4)



c) x=9, 9 ≠ 9.p

6 = 9.p = find-set(6)

x=6 6.p = find-set(6.p)

6.p = find-set(4)

x=4, 4 ≠ 4.p so find-set(4.p)

4.p = find-set(0)

x=0, 0 ≠ 0.p is FALSE

so return 0.*

Path is 9 → 6 → 4 → 0

9.p points to 0 finally.

6.p points to 0

4.p points to 0 now

* 9.p, 6.p, 4.p all now point to 0.